

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W002
Well Name	SYN-Assam-Duliajan-002
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.389232 N
Longitude	95.288712 E
Well Status	Abandoned
Well Role	Historical (offset reference well)
Spud Date	12-Feb-2016
Drilling Completed	23-Feb-2016
Report / Completion Date	03-Mar-2016
Total Depth (Driller)	3,450 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-04
Drilling Days	12 days
Recorded Events	7
Recorded NPT	94.3 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W002
Well Name: SYN-Assam-Duliajan-002
Field: SYN-Assam-Duliajan
Latitude: 27.389232
Longitude: 95.288712
Total Depth: 3,450 m
Primary Formation: Sandstone
Well Status: Abandoned
Spud Date: 2016-02-12
Event Count: 7
Total NPT: 94.3 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

DOC No. WCR-W002-R0 · ISSUE 03-Mar-2016 · PROTOTYPE TRAINING RECORD

3. WELL SUMMARY

3.1 Objective

Well W002 (SYN-Assam-Duliajan-002) was drilled to test the deeper sandstone/limestone sequence and record drilling behaviour across the intermediate shale section. The well was drilled vertically from surface to a driller's total depth of 3,450 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,450 m, completed in 12 rotary drilling days between 12-Feb-2016 and 23-Feb-2016. Average overall penetration rate for the well was 12.0 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–785 m), Shale (785–1,880 m), Sandstone (1,880–3,083 m), Limestone (3,083–3,450 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 7 reportable events were recorded, comprising 2 × Mud Loss, 1 × Wellbore Instability, 2 × Stuck Pipe, 1 × Cementing Issue, 1 × Kick. Aggregate non-productive time was 94.3 hours. Event detail is given in Section 6.

3.5 Casing Operations

3 casing strings were run: 13-3/8" at 785 m, 9-5/8" at 1,880 m, 7" at 3,450 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Abandoned**. The well was plugged and abandoned after evaluation; cement plugs were set across the open hole and shoe intervals.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W005	SYN-Assam-Duliajan-005	4.61	3,560	Sandstone	5
W001	SYN-Assam-Duliajan-001	4.70	3,470	Sandstone	4
W007	SYN-Assam-Duliajan-007	4.73	3,400	Sandstone	3
W010	SYN-Assam-Duliajan-010	5.00	3,260	Sandstone	7
W004	SYN-Assam-Duliajan-004	6.54	3,590	Sandstone	2
W003	SYN-Assam-Duliajan-003	7.85	3,850	Sandstone	8

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	785	785	Mud Loss recorded (1 event)
2	Shale	785	1,880	1,095	Wellbore Instability, Stuck Pipe, Cementing Issue recorded (4 events)
3	Sandstone	1,880	3,083	1,203	Mud Loss recorded (1 event)
4	Limestone	3,083	3,450	367	Kick recorded (1 event)

Formation: Claystone Top: 0 m Bottom: 785 m
Formation: Shale Top: 785 m Bottom: 1880 m
Formation: Sandstone Top: 1880 m Bottom: 3083 m
Formation: Limestone Top: 3083 m Bottom: 3450 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–785	9.56	9.0–10.2	31.8	141	8.0
Shale	785–1,880	11.00	10.2–11.8	16.5	125	17.3
Sandstone	1,880–3,083	12.42	11.6–13.3	15.7	113	25.1
Limestone	3,083–3,450	13.90	13.7–14.2	7.7	91	34.6

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 13.9 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 121 rpm range on average with weight on bit averaging 19.8 klbf. Penetration rate declined with depth from 31.8 m/hr in the upper hole to 7.7 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0021	476	Claystone	Mud Loss	Low	2.2
E0026	1,098	Shale	Wellbore Instability	High	26.1
E0022	1,128	Shale	Stuck Pipe	Medium	11.5
E0023	1,401	Shale	Cementing Issue	High	20.5
E0024	1,506	Shale	Stuck Pipe	Low	5.5
E0002	2,800	Sandstone	Mud Loss	Critical	18.7
E0025	3,307	Limestone	Kick	Medium	9.8

Well ID: W002		Event Type: Mud Loss		Depth: 476 m		Formation: Claystone		Severity: Low		Duration: 2.2 hours
Well ID: W002		Event Type: Wellbore Instability		Depth: 1098 m		Formation: Shale		Severity: High		Duration: 26.1 hours
Well ID: W002		Event Type: Stuck Pipe		Depth: 1128 m		Formation: Shale		Severity: Medium		Duration: 11.5 hours
Well ID: W002		Event Type: Cementing Issue		Depth: 1401 m		Formation: Shale		Severity: High		Duration: 20.5 hours
Well ID: W002		Event Type: Stuck Pipe		Depth: 1506 m		Formation: Shale		Severity: Low		Duration: 5.5 hours
Well ID: W002		Event Type: Mud Loss		Depth: 2800 m		Formation: Sandstone		Severity: Critical		Duration: 18.7 hours
Well ID: W002		Event Type: Kick		Depth: 3307 m		Formation: Limestone		Severity: Medium		Duration: 9.8 hours

7.1 EVENT DETAIL (1–3 OF 7)

EVENT E0021 — MUD LOSS AT 476 m (CLAYSTONE)

Well ID: W002
Event ID: E0021
Event Type: Mud Loss
Depth: 476 m
Formation: Claystone
Severity: Low
Duration: 2.2 hours
Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment.

Description. Partial loss of drilling fluid recorded at 476 m in the Claystone interval; returns reduced and LCM pumped to regain circulation. Parameters recorded around the event depth: mud weight 9.66 ppg, ROP 31.8 m/hr, 140 rpm, WOB 8.8 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

EVENT E0026 — WELLBORE INSTABILITY AT 1,098 m (SHALE)

Well ID: W002
Event ID: E0026
Event Type: Wellbore Instability
Depth: 1098 m
Formation: Shale
Severity: High
Duration: 26.1 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Sloughing Shale at 1098 m produced excess cuttings volume and hole enlargement; mud properties adjusted. Parameters recorded around the event depth: mud weight 10.66 ppg, ROP 16.8 m/hr, 125 rpm, WOB 15.8 klbf.

Response. Mud inhibition and rheology increased, sweeps circulated to clear cavings and controlled back-reaming carried out; ROP limited through the unstable interval.

Outcome. Hole condition stabilised and cuttings volume returned to normal. Interval drilled to section TD without further deterioration.

Historical observation. Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.

EVENT E0022 — STUCK PIPE AT 1,128 m (SHALE)

Well ID: W002
Event ID: E0022
Event Type: Stuck Pipe
Depth: 1128 m
Formation: Shale
Severity: Medium
Duration: 11.5 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Drill string became stuck at 1128 m within the Shale section; overpull observed and jarring / working of pipe required. Parameters recorded around the event depth: mud weight 10.68 ppg, ROP 15.7 m/hr, 122 rpm, WOB 16.2 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

7.2 EVENT DETAIL (4–6 OF 7)

EVENT E0023 — CEMENTING ISSUE AT 1,401 m (SHALE)

Well ID: W002
Event ID: E0023
Event Type: Cementing Issue
Depth: 1401 m
Formation: Shale
Severity: High
Duration: 20.5 hours
Operational Impact: Significant non-productive time; remedial operations and additional materials required.

Description. Cement slurry losses while cementing across Shale at 1401 m; top of cement lower than designed. Parameters recorded around the event depth: mud weight 11.08 ppg, ROP 17.2 m/hr, 126 rpm, WOB 17.6 klbf.

Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage.

Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage.

Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

EVENT E0024 — STUCK PIPE AT 1,506 m (SHALE)

Well ID: W002
Event ID: E0024
Event Type: Stuck Pipe
Depth: 1506 m
Formation: Shale
Severity: Low
Duration: 5.5 hours
Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment.

Description. Drill string became stuck at 1506 m within the Shale section; overpull observed and jarring / working of pipe required. Parameters recorded around the event depth: mud weight 11.22 ppg, ROP 15.0 m/hr, 117 rpm, WOB 17.6 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

EVENT E0002 — MUD LOSS AT 2,800 m (SANDSTONE)

Well ID: W002
Event ID: E0002
Event Type: Mud Loss
Depth: 2800 m
Formation: Sandstone
Severity: Critical
Duration: 18.7 hours
Operational Impact: Major non-productive time; section objectives affected and specialist intervention required.

Description. Seepage-to-partial mud losses observed while drilling Sandstone at 2800 m; mud weight trimmed and losses monitored. Parameters recorded around the event depth: mud weight 12.73 ppg, ROP 16.7 m/hr, 106 rpm, WOB 25.7 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

7.3 EVENT DETAIL (7–7 OF 7)

EVENT E0025 — KICK AT 3,307 m (LIMESTONE)
Well ID: W002 Event ID: E0025 Event Type: Kick Depth: 3307 m Formation: Limestone Severity: Medium Duration: 9.8 hours Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned. Description. Unexpected formation influx in Limestone at 3307 m; flow check positive, BOP closed and circulation of kill mud carried out. Parameters recorded around the event depth: mud weight 13.89 ppg, ROP 7.6 m/hr, 88 rpm, WOB 35.3 kbf. Response. Flow check positive; well shut in on the annular preventer, SIDPP/SICP recorded and kill mud circulated using the driller's method; mud weight raised and the well monitored on the trip tank. Outcome. Well killed with no further influx. Kill mud weight retained for the remainder of the section and flow checks made on every connection. Historical observation. A permeable, slightly over-pressured interval was proven; trip margin and flow-check discipline are the controlling factors.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	785	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	1,880	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 670 m
7"	N-80, 29 lb/ft	3,450	8-1/2"	Tail 15.8 ppg, TOC 1,730 m

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0023** — Cementing Issue at 1,401 m in Shale (High, 20.5 hr). Cement slurry losses while cementing across Shale at 1401 m; top of cement lower than designed.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,450 m
Primary Formation	Sandstone
Major Events	3 of 7 events rated High or Critical
Total NPT	94.3 hours
Completion Status	Abandoned
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Mud Loss** at 476 m, 2,800 m (Claystone, Sandstone) — 20.9 hr cumulative.
- **Wellbore Instability** at 1,098 m (Shale) — 26.1 hr cumulative.
- **Stuck Pipe** at 1,128 m, 1,506 m (Shale) — 17.0 hr cumulative.
- **Cementing Issue** at 1,401 m (Shale) — 20.5 hr cumulative.
- **Kick** at 3,307 m (Limestone) — 9.8 hr cumulative.

9.2 Key Historical Observations

- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Reactive/laminated section sensitive to mud chemistry and exposure time; cavings volume is the earliest indicator.
- Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.
- Annular coverage is sensitive to losses and centralisation across the shoe interval.
- A permeable, slightly over-pressured interval was proven; trip margin and flow-check discipline are the controlling factors.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Approach the Claystone/Sandstone interval (476–2,800 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.
- Raise inhibition before entering the Shale section and limit open-hole exposure time; treat rising cavings volume as an early warning.
- Improve hole cleaning and minimise stationary time between 1,128 and 1,506 m; plan a wiper trip before extended surveys in this interval.
- Review slurry design, centralisation and displacement rate for the stage covering 1,401 m; run a bond log before proceeding.
- Maintain trip margin and flow-check discipline approaching 3,307 m; the interval is capable of influx and should be drilled with kill-sheet updated.

10. DOCUMENT CONTROL

Document Number	WCR-W002-R0
Revision	R0
Issue Date	03-Mar-2016
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formations.csv, drilling_events.csv, drilling_parameters.csv